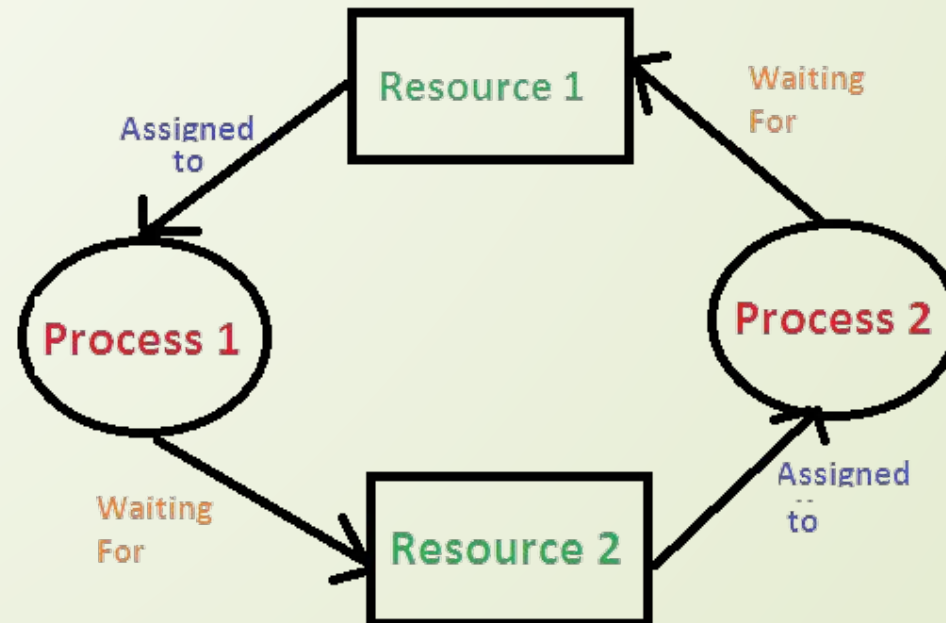




Models of Deadlocks

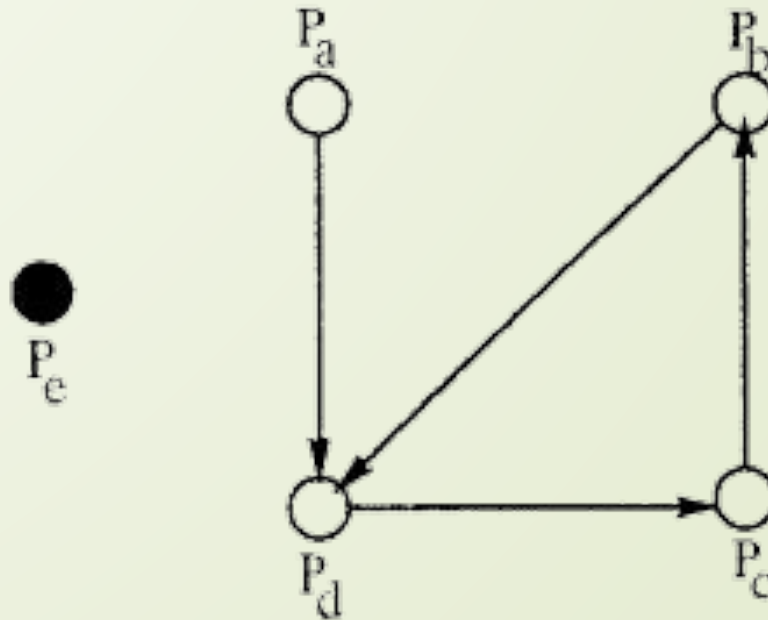
- Deadlocks in distributed systems occur when processes are unable to proceed due to circular dependencies on resources or messages held by other processes.
- Distributed deadlocks can arise due to the nature of communication and coordination between processes over a network.



Single Resource Model

- A process can have at most one outstanding request for only one unit of a resource.
- The maximum out-degree of a node in a WFG can be 1.
- The presence of a cycle in the WFG shall indicate that there is a deadlock.

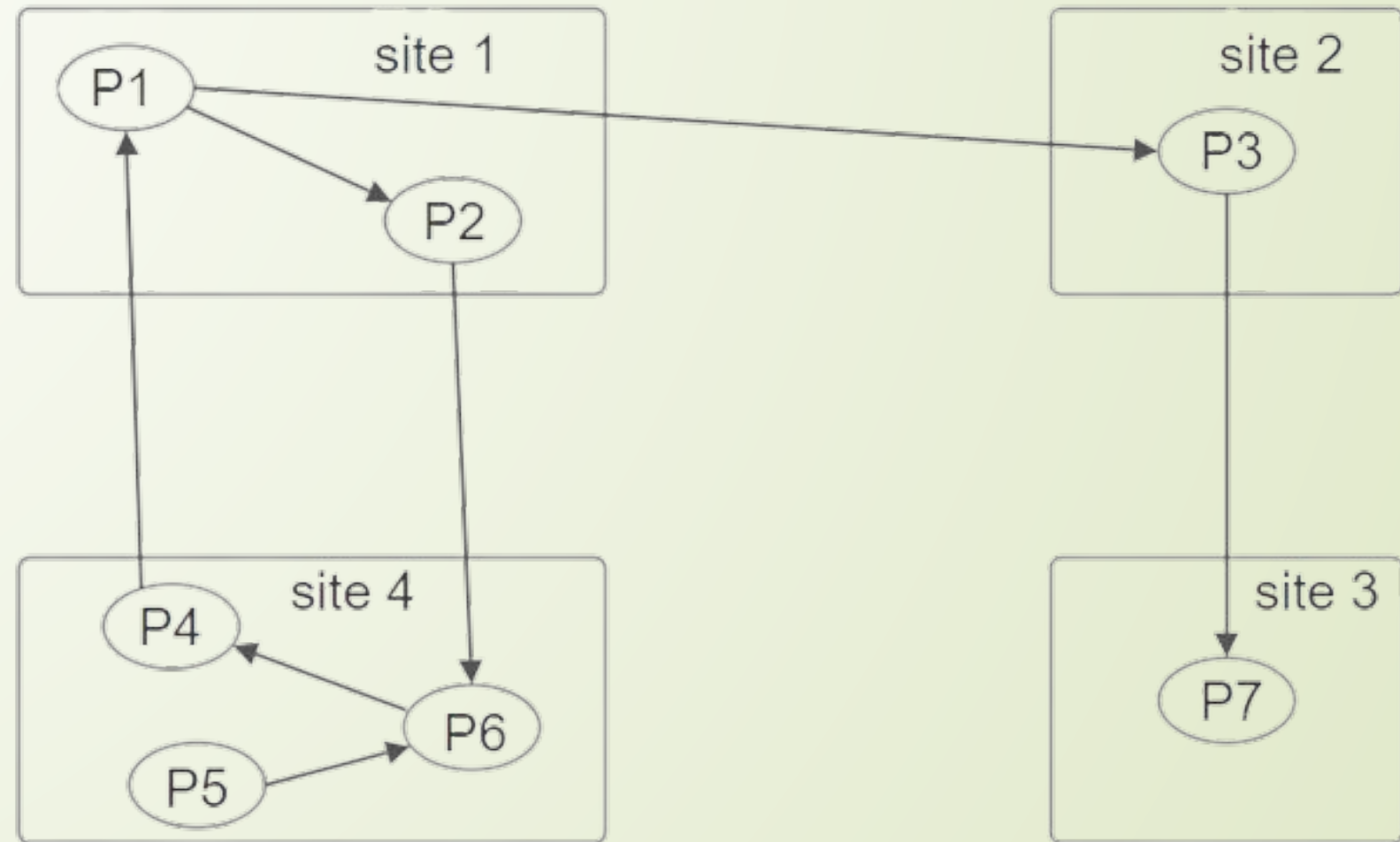
Example:



AND Model

- an idle process becomes active only after all the grant messages are received.
- a process can request more than one resource simultaneously.
- request is satisfied only after all requested resources are granted.
- The requested resources may exist at different locations.
- Out degree of a node(AND nodes) in the WFG can be more than 1.
- presence of a cycle in the WFG indicates a deadlock.
- if a cycle is detected then a deadlock is there but not vice versa.

Example:



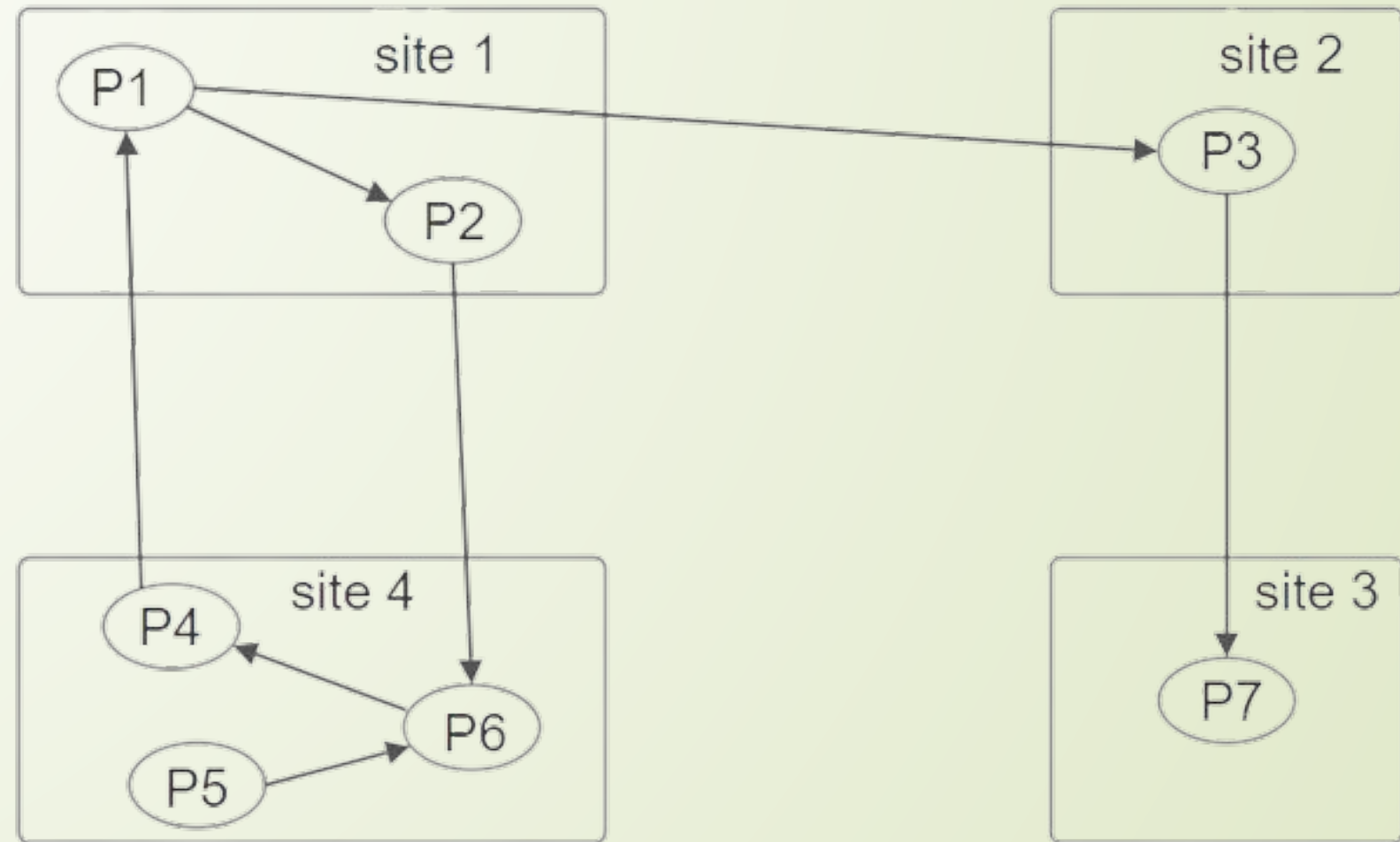
OR Model

- A grant message from any of the processes in a dependent set can make an idle process active.
- a process can request more than one resource simultaneously (out degree > 1)
- a process P_i is blocked if it has a pending OR request to be satisfied.
- A process is permanently blocked if it never gets a grant message.
- a cycle in the WFG of an OR model does not imply a deadlock.
- The presence of a knot indicates a deadlock.

OR Model

- For every blocked process, there is an associated set of processes called dependent set.
- Dependent set should be the subset of whole set(S).
- In short, a process is deadlocked or permanently blocked, if the following conditions are met:
 - Each of the process in the set S is blocked.
 - The dependent set for each process in S is a subset of S.
 - No grant message is in transit between any two processes in set S.

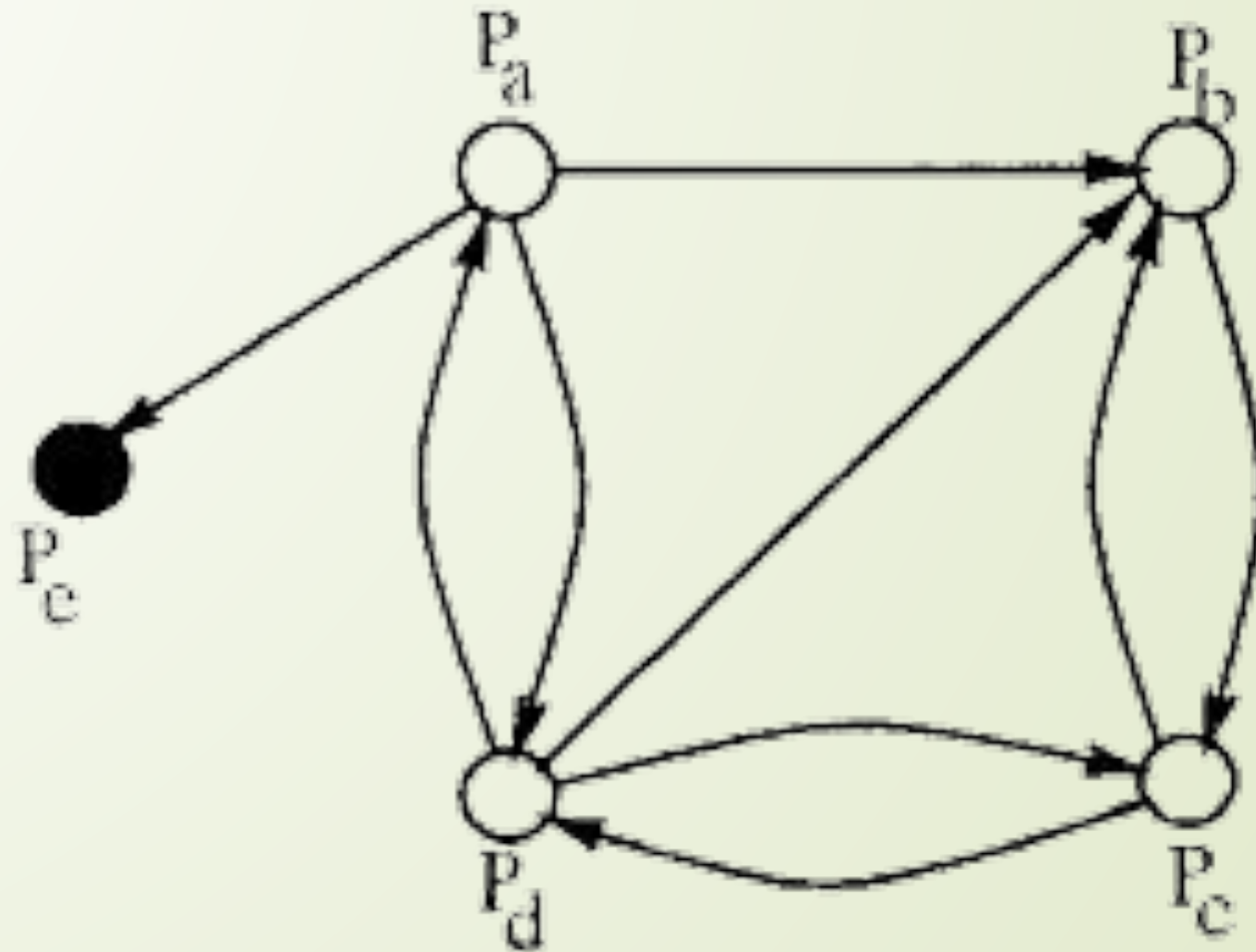
Example:



(P_q) Model (p out of q model)

- This is a variation of AND-OR model.
- This allows a request to obtain any k available resources from a pool of n resources. Both the models are the same in expressive power.
- This favours more compact formation of a request.
- Every request in this model can be expressed in the AND-OR model and vice-versa.
- AND requests for p resources can be stated as (P_q) and OR requests for p resources can be stated as (P_1) .

Example:





Unrestricted model

- No assumptions are made regarding the underlying structure of resource requests.
- In this model, only one assumption that the deadlock is stable is made and hence it is the most general model.
- This model helps separate concerns: Concerns about properties of the problem (stability and deadlock) are separated from underlying distributed systems computations (e.g., message passing versus synchronous communication).